🤖 Pinterest AI Content Generation System

This is a comprehensive 3-stage automation system designed to generate and prepare content for the Pinterest bulk uploader using the Gemini AI and Google Drive.

This system will:

1. **Generate Content:** Create unique image prompts, titles, descriptions, and keywords based on product and style data.
2. **Generate Images:** Use the prompts from Step 1 to automatically generate unique product images with Gemini.
3. **Upload & Format:** Upload the generated images to Google Drive and format all data into CSV files ready for the Pinterest bulk uploader.

🌟 Key Features

- **AI Content Generation:** Automatically writes titles, descriptions, image prompts, and keywords.
- **AI Image Generation:** Creates beautiful, Pinterest-ready images based on prompts and reference product photos.
- **Google Drive Integration:** Automatically uploads generated images to Google Drive and retrieves shareable links.
- **Bulk CSV Export:** Generates perfectly formatted CSV files for the Pinterest bulk uploader.
- **Dynamic Validation:** Interactively asks whether certain fields (like `Affiliate_Link`) should be mandatory.
- **Error Handling:** Intelligently separates *input data errors* (like missing images) from *API generation errors* (like failed API calls) into separate files for easy review.
- **Duplicate Prevention:** Automatically detects duplicate titles during generation and attempts to rewrite them to ensure content uniqueness.
- **Template Customization:** Uses external `.txt` files to customize the AI prompts for content and images.

⚙️ Workflow & Order of Operations

This system is designed to be run in a specific 3-step sequence. **You must run the scripts in this order.**

➡️ Step 1: `Pinterest_Gen_System01_content.py` (Content Generation)

This script reads your product and style Excel files and uses the Gemini AI to generate text content for each product-style combination.

- When run, this script will interactively ask for:
  - `test_single` mode (Y/N)
  - `verbose_debug` output (Y/N)
  - Whether `Affiliate_Link` is a required field (Y/N)
- It automatically detects and attempts to regenerate duplicate titles.
- **Inputs:** `Pinterest_Affiliate_Data.xlsx`, `Pinterest_Affiliate_Product.xlsx`, `product_image/` (original images), `image_prompt_template.txt`, `content_prompt_template.txt`.
- **Outputs:**
  - `Generated_Content/Generated_Content_[timestamp].xlsx`: An Excel file containing all successfully generated data. API errors are saved to a separate sheet named `Generation_Errors` within this file.
  - `Input_Errors_[timestamp].xlsx`: (In the root folder) A separate Excel file containing any products that failed *input validation* (e.g., missing product data, image file not found).

➡️ Step 2: `Pinterest_Gen_System02_image.py` (Image Generation)

This script reads the `Generated_Content...xlsx` file from Step 1. It uses the `Image_Prompt_For_AI` column and the *original* product image to generate a new, stylized image with the Gemini AI.

- **Inputs:** `Generated_Content/Generated_Content_[timestamp].xlsx` (from Step 1), `product_image/` (original images), `image_generation_instruction.txt`.
- **Outputs:**
  - Generated `.png` images are saved into a new subfolder: `Generated_Content/[Excel_Name_Subfolder]/` (e.g., `Generated_Content/Generated_Content_202511151500/`).

➡️ Step 3: `Pinterest_Gen_System03_image_upload.py` (Upload & CSV Generation)

This script reads the Excel file from Step 1 and the generated images from Step 2. It uploads each generated image to Google Drive, gets a shareable link, and formats all the data into the final CSV files for Pinterest.

- **Inputs:** `Generated_Content/Generated_Content_[timestamp].xlsx` (from Step 1), `Generated_Content/[Excel_Name_Subfolder]/` (images from Step 2), `board_mapping.txt`, `drive_config.txt`, `client_secret.json`.
- **Outputs:**
    - `pinterest_bulk_upload_[timestamp]_[board_name].csv`: The final, ready-to-upload CSV files, split by Pinterest board.
    - `ERROR_ROWS_[timestamp].csv`: Any rows that failed final validation (e.g., missing generated image, no board mapping).
    - `token.json`: (Auto-generated) Stores your Google Drive authentication.

📂 Required File Structure

/Your_Project_Folder
│
├── 📜 Pinterest_Gen_System01_content.py
├── 🎨 Pinterest_Gen_System02_image.py
├── 📤 Pinterest_Gen_System03_image_upload.py
│
├── 🔑 .env (Contains GEMINI_API_KEY)
├── 🔑 client_secret.json (For Google Drive API)
│
├── 📊 Pinterest_Affiliate_Data.xlsx (Input)
├── 📦 Pinterest_Affiliate_Product.xlsx (Input)
│
├── ⚙️ board_mapping.txt (Auto-created / Must Edit)
├── ⚙️ drive_config.txt (Auto-created / Must Edit)
│
├── 📝 content_prompt_template.txt (Auto-created / Can Edit)
├── 📝 image_generation_instruction.txt (Auto-created / Can Edit)
├── 📝 image_prompt_template.txt (Auto-created / Can Edit)
│
├── 🖼️ /product_image (Input)
│ └── original_image_01.jpg
│ └── original_image_02.png
│
├── 🚀 /Generated_Content (Output)
│ ├── Generated_Content_202511151500.xlsx (From Step 1)
│ └── /Generated_Content_202511151500 (From Step 2)
│ └── ProductID_StyleName.png
│
├── ⚠️ Input_Errors_...xlsx (Output from Step 1)

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├── ⚠️ ERROR_ROWS_...csv (Output from Step 3)
├── ✅ pinterest_bulk_upload_...csv (Output from Step 3)
└── 🔒 token.json (Auto-generated by Step 3)
```

🛠️ Setup and Configuration

1. Python Environment

Install all required libraries:

Bash
pip install pandas google-genai openpyxl python-dotenv Pillow jinja2 google-auth google-auth-oauthlib google-auth-httplib2 google-api-python-client

2. Google Gemini API Key Setup

1. Create an API Key in the Google AI Studio.
2. **Enable Billing:** This is a **mandatory step** for this system.
    - While your API key grants access, Google provides different free quotas for different models.
    - The model used in **System 01 (Text)** may work on a basic free tier.
    - However, the model used in **System 02 (Image Generation)** (e.g., `gemini-2.5-flash-image`) **requires billing to be enabled**. The free tier limit for this model is `0`, and you will receive a `Quota exceeded` error if billing is not active.
3. **Free Credit:** Enabling billing allows Google to charge for usage, but it also activates your $300 free credit (if your account is new).
4. Create a file named `.env` in the root project folder.
5. Add your single API key to the `.env` file. This one key is used for all models in this project.
   GEMINI_API_KEY=Your_API_Key_Here

3. Google Drive API (For Step 3)

You need OAuth 2.0 credentials to allow the script to upload files to your Google Drive.

1. **Enable API:** Go to the Google Cloud Console and enable the "Google Drive API" for your project.
2. **Configure OAuth Consent Screen:**
    - Go to "APIs & Services" > "OAuth consent screen".

    - Select "External" and fill in the required app name, user support email, etc.
3. **Create Credentials:**
    - Go to "Credentials" > "+ CREATE CREDENTIALS" > "OAuth client ID".
    - Select "Desktop App" as the Application type.
    - Give it a name and click "Create".
4. **Download JSON:**
    - After creation, click the "DOWNLOAD JSON" button.
    - **Rename the downloaded file to `client_secret.json`** and place it in your root project folder.
5. **Create Google Drive Folder:**
    - Create a folder in your Google Drive where you want the images to be uploaded.
    - Copy the **Folder ID** from the URL (e.g., `.../folders/THIS_IS_THE_ID`).
6. **Run & Authenticate:**
    - The first time you run `Pinterest_Gen_System03_image_upload.py`, it will:
        - Create `drive_config.txt` and `board_mapping.txt` if they don't exist.
        - Open a browser window for you to log in and authorize Google Drive access.
        - Save your credentials in `token.json` for future use.
    - **Important:** Open the created `drive_config.txt` and paste your Google Drive Folder ID into it before running the script again.

4. Input Data Files

- **`Pinterest_Affiliate_Data.xlsx`:**
    - `Model_Config`: Configure the Gemini model to use.
    - `Styles_Config`: Define your `Style_ID` and `style_Name`.
    - `Process_Config`: List which sheets from `Pinterest_Affiliate_Product.xlsx` to process.
- **`Pinterest_Affiliate_Product.xlsx`:**
    - Must contain sheets with names that match the entries in the `Process_Config` sheet.
    - **Required Columns:** `Product_ID`, `Product_Name`, `Image_name`, `Product_desc`. (Note: The script will automatically rename a column named `Pinterest_Angle` to `Product_desc` if it exists).
    - **Conditionally Required Column:** `Affiliate_Link`. When you run `Pinterest_Gen_System01_content.py`, you will be asked if this field is required. If you select 'Y', products missing this link will be sent to the `Input_Errors` file. If you select 'N', it will be treated as an optional field.
- **`/product_image`:**

    - Place all your *original* product images here. The filenames must match the `Image_name` column in `Pinterest_Affiliate_Product.xlsx`.

5. Template & Config Files

The scripts will automatically create these `.txt` files with default content if they are missing. You can edit them to change the AI's behavior.

- **`image_prompt_template.txt` (Step 1):** Controls how the AI generates the *image prompt* (the prompt used to create the image in Step 2).
- **`content_prompt_template.txt` (Step 1):** Controls how the AI generates the text content.
    - **WARNING:** Do not change the `[TITLE]`, `[DESCRIPTION]`, and `[HASHTAGS]` tags. They are required by the script to parse the AI's output.
- **`image_generation_instruction.txt` (Step 2):** Provides general instructions (e.g., style, lighting) when the AI *generates* the actual image.
- **`board_mapping.txt` (Step 3):**
    - **This file MUST be edited manually.**
    - Maps the `Style_ID` from your Excel file to the exact, case-sensitive Pinterest board name.
    - Format: `Style_ID : Pinterest_Board_Name` (e.g., `1 : Cozy Living Room`)
- **`drive_config.txt` (Step 3):**
    - **This file MUST be edited manually.**
    - Place your Google Drive Folder ID here.